

Nikhil Shirsathe

Pune, Maharashtra

+91 9075543862 — Nikhilshirsathe@gmail.com

LinkedIn : www.linkedin.com/in/nikhil-shirsathe

GitHub : github.com/Nikhilshirsathe

Huggingface : huggingface.co/Nikss2709

Profile Summary

AI & Machine Learning undergraduate specializing in Natural Language Processing and Computer Vision, with a strong interest in research-driven problem solving and applied AI systems. Experience working across diverse AI tasks including transformer-based NLP, document understanding, and real-world data pipelines, with a focus on experimentation, evaluation, and iterative improvement. Smart India Hackathon 2025 Winner (Team Lead) and AI/ML Intern at NTRO, with hands-on exposure to building and analyzing scalable ML solutions and exploring open-ended problems in practical settings.

Experience

AI/ML Intern — BAU AI, Germany (Remote) 15 June 2026 – Present

- Working remotely with a Germany-based AI company to develop intelligent AI solutions for real-world applications.
- Designing, developing, and optimizing machine learning models and AI pipelines using modern AI frameworks and industry best practices.

AI/ML Intern — NTRO, New Delhi 20 Jan 2026 - 20 April 2026 (3 Months)

- Worked on a multilingual NLP system (Sanskriti) using transformer-based architectures for South Asian languages.
- Contributed to model development, optimization, and evaluation for translation tasks.
- Built preprocessing pipelines and integrated OpenCV-based text extraction with document structure retention for complex document inputs.
- Improved model performance through experimentation, inference optimization, and deployment in production-like environments.

AI/ML Intern — Mixins Technology, Pune Aug 2025 – Nov 2025 (3 Months)

- Built end-to-end ML pipelines including preprocessing, feature engineering, and evaluation.
- Applied data science techniques to improve model performance on real-world datasets.
- Performed exploratory data analysis and developed predictive models using Python tools.

Projects

Sanskriti — SIH 2025 Winning Project (Transformer-Based NLP System)

- Led a team to develop a transformer-based multilingual translation system using Hugging Face for low-resource languages.
- Designed a hybrid NLP pipeline with sequence-to-sequence models and integrated VLM-based text extraction with document structure retention.
- Optimized inference pipelines for real-time performance and improved translation accuracy through model tuning.
- Achieved national-level recognition as Winner of Smart India Hackathon 2025.

MediQ — AI-Powered EMR with Agentic Workflow

- Built an AI-powered EMR system integrating agentic workflows for intelligent healthcare data processing.
- Implemented speech-to-text and healthcare Named Entity Recognition (NER) pipelines for extracting structured medical information.
- Integrated Google Vision AI for document analysis and LLMs for contextual clinical insights.
- Designed multimodal pipelines combining speech, text, and document data for end-to-end automation.

Hackathons & Achievements

Smart India Hackathon (SIH) 2025 — Winner (Team Leader)

Technical Skills

Programming Languages :

Python, SQL, C/C++

Machine Learning & Data Science :

Scikit-learn, Pandas, NumPy, Feature Engineering, Model Evaluation and Deployment

Natural Language Processing (NLP) :

Hugging Face Transformers, Tokenization, Sequence-to-Sequence Models

Computer Vision :

OpenCV, Image Processing, OCR Pipelines, Document Image Analysis

Deep Learning :

PyTorch, Model Fine-tuning, LoRA (Low-Rank Adaptation)

AI Systems & Workflows :

LLM-based pipelines, Prompt Design, Multimodal Integration

Education

B.Tech in Artificial Intelligence & Machine Learning

MIT Academy of Engineering, Pune

CGPA : 8.23/10

Research Interests

Natural Language Processing (Transformers, LLMs)

Computer Vision (OpenCV, Image Analysis)

Applied AI & Model Optimization

Agentic AI